Google Cloud H2S

Gen AI Academy

APAC Edition

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Participant Details

a. Participant name: **Vijay Dwivedi**
b. Problem Statement: **ShikshaAI - An Intelligent Educational Agent for Lesson Generation and Academic Q&A**

Brief about the idea

ShikshaAI is a classroom-ready AI teaching assistant that helps educators generate structured lesson modules instantly. Teachers simply provide the subject, grade, topic, and preferred language, and ShikshaAI delivers a complete lesson package including an age-appropriate explanation, a classroom activity, and three homework questions. It also answers general academic questions, making it a versatile companion for both lesson planning and in-class support.

Link for GitHub repo: https://github.com/VijayDwivedi-ml/shikshaai

Meeting the Build Criteria

- How did you navigate the requirements of your chosen track to build a custom solution - using ADK, MCP, or AlloyDB AI that effectively addresses the 'What You Must Build' criteria?

The solution was designed to strictly align with the Track 1 requirement of building a single AI agent using ADK and Gemini, deployed via an HTTP endpoint on Cloud Run.

A single agent (ShikshaAI) was implemented using the ADK framework and powered by a Gemini model for inference.

The core capability of the agent is **intent classification and request routing**, where user inputs are analyzed and directed to fixed response logic such as structured lesson generation or academic question answering.

The agent accepts input via an HTTP endpoint and returns a valid, structured response, thereby fulfilling the requirement of building a deployable, callable AI agent with a clearly defined functional behavior.

Opportunities

- How different is it from any of the other existing ideas?
- How will it be able to solve the problem?
- USP of the proposed solution

Unlike many single-purpose agents that perform only one static task (such as summarization or Q&A), ShikshaAI introduces a **routing-based agent design** that dynamically interprets user intent and directs it to appropriate response logic within a single unified interface.

This approach improves usability by eliminating the need for multiple tools, allowing users to interact naturally while the agent handles intent detection and response selection. It is particularly useful in educational contexts where user inputs can vary widely in structure and intent.

The key USP of the solution is its **intent-driven routing mechanism combined with structured output generation**, enabling a balance between flexibility (handling different input types) and control (ensuring consistent, well-formatted responses such as lesson modules).

List of features offered by the solution

Intent Classification Engine : Automatically interprets user input to identify whether it is a greeting, lesson request, or academic question.

Dynamic Request Routing : Routes user requests to predefined response logic based on detected intent, ensuring appropriate handling.

Structured Lesson Module Generation : Generates well-formatted lesson content including explanation, classroom activity, and exactly three homework questions.

Academic Question Answering : Provides clear, concise, and teacher-like explanations for general academic queries.

Input Validation & Guidance : Detects incomplete lesson requests and prompts users to provide missing details (Subject, Grade, Topic, Language).

Multi-language Support : Generates lesson content in the user-specified language.

Consistent Output Formatting : Ensures standardized responses with clear sections for better readability and usability.

HTTP Endpoint Accessibility : Exposes the agent via an API endpoint, enabling easy integration and deployment on Cloud Run.

Process flow diagram or Use-case diagram

The system processes user input through an API layer, which invokes the ADK-based agent. The agent performs intent classification and routes the request through a decision layer to predefined response logic, such as greeting, lesson generation, or question answering. The output is then formatted into a structured response and returned to the user.

```
User Request (HTTP Input)
        ↓
API Layer
        ↓
AI Agent (ADK-based)
        ↓
Intent Classification
        ↓
Decision & Routing Layer
        ↓
----------------------------------------
|                |                      |
↓                ↓                      ↓
Greeting Logic   Lesson Generation      Q&A Logic
                 Logic                  (Answer Generation)
        ↓
Structured Response Generation
        ↓
HTTP Response Output
```

Technologies to be used in the solution

Google ADK (Agent Development Kit) – Agent definition and orchestration

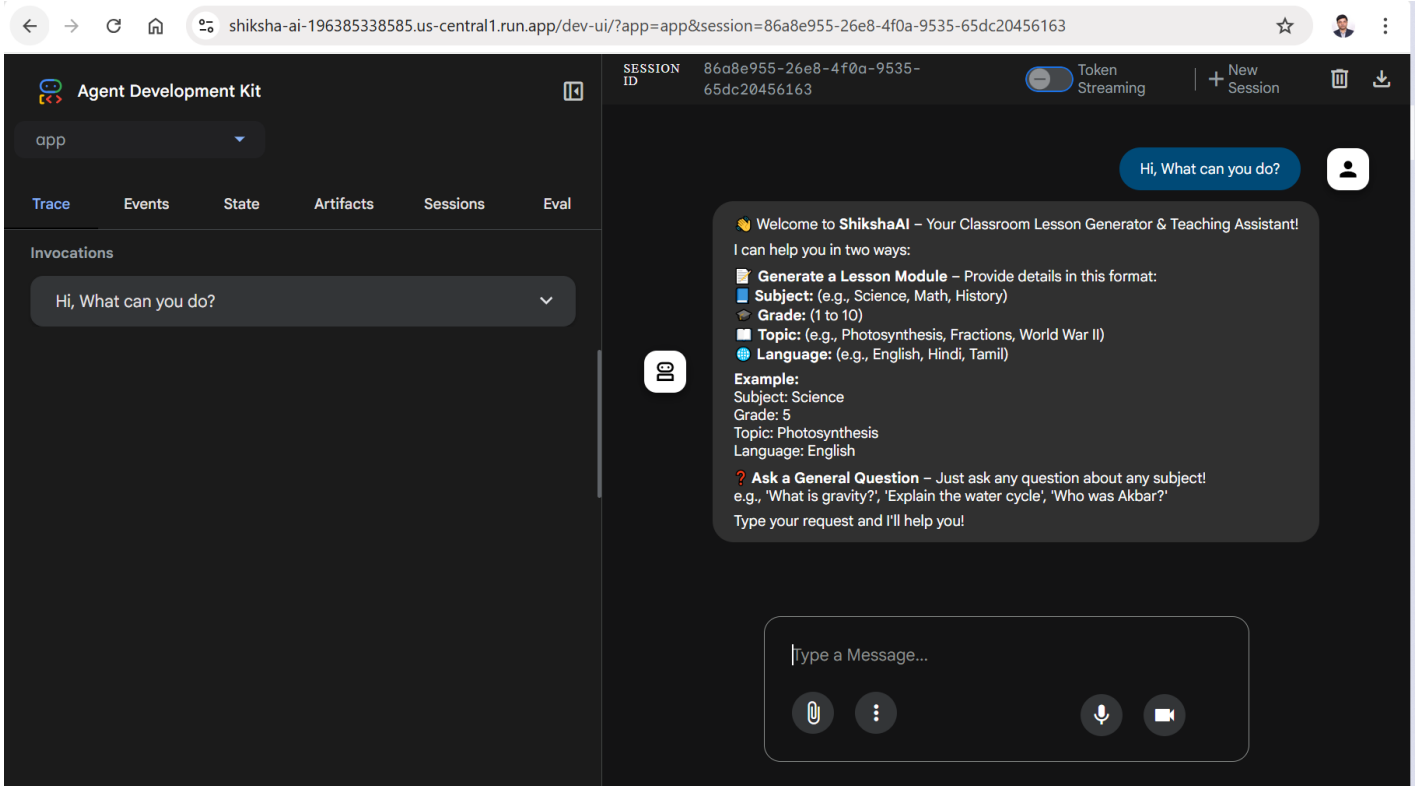
Gemini Model – Natural language processing and inference

Python – Core application logic

HTTP API Layer – Input/output interface for the agent

Docker – Containerization and deployment

Snapshots of the prototype (1 of 3)



Snapshots of the prototype (2 of 3)

shiksha-ai-196385338585.us-central1.run.app/dev-ui/?app=app&session=86a8e955-26e8-4f0a-9535-65dc20456163

SESSION ID 86a8e955-26e8-4f0a-9535-65dc20456163

Token Streaming | + New Session

Hi, What can you do?

👋 Welcome to **ShikshaAI** – Your Classroom Lesson Generator & Teaching Assistant!

I can help you in two ways:

📝 **Generate a Lesson Module** – Provide details in this format:
📘 **Subject:** (e.g., Science, Math, History)
🎓 **Grade:** (1 to 10)
📖 **Topic:** (e.g., Photosynthesis, Fractions, World War II)
🌐 **Language:** (e.g., English, Hindi, Tamil)

Example:
Subject: Science
Grade: 5
Topic: Photosynthesis
Language: English

❓ **Ask a General Question** – Just ask any question about any subject!
e.g., 'What is gravity?', 'Explain the water cycle', 'Who was Akbar?'

Type your request and I'll help you!

Subject: Maths
Grade: 10
Topic: Quadratic equation
Language: English

Snapshots of the prototype (3 of 3)

shiksha-ai-196385338585.us-central1.run.app/dev-ui/?app=app&session=86a8e955-26e8-4f0a-9535-65dc20456163

SESSION ID 86a8e955-26e8-4f0a-9535-65dc20456163

Token Streaming | + New Session

Language: English

Here is a lesson module on Quadratic Equations for Grade 10 students:

Explanation

A **quadratic equation** is a special type of algebraic equation that involves a variable raised to the power of 2, but no higher power. It's a polynomial equation of the second degree. The standard form of a quadratic equation is:

$ax^2 + bx + c = 0$

where:

- 'x' represents the unknown variable.
- 'a', 'b', and 'c' are constants (numbers).
- 'a' cannot be zero (if a=0, it becomes a linear equation, not quadratic).
- 'a', 'b', and 'c' are called coefficients.

The solutions to a quadratic equation are also known as its **roots** or **zeros**. A quadratic equation can have up to two distinct real roots, one repeated real root, or two complex roots.

There are several methods to find the roots of a quadratic equation:

1. **Factoring:** Breaking down the quadratic expression into a product of two linear factors.
2. **Completing the Square:** Transforming the equation into a perfect square trinomial.
3. **Quadratic Formula:** A universal formula that always provides the roots:

   $x = [-b \pm \sqrt{(b^2 - 4ac)}] / 2a$

   This formula is derived from the completing the square method and is very powerful because it works for any quadratic equation, regardless of whether it can be factored easily or not. The term **$(b^2 - 4ac)$** is called the **discriminant**, which tells us about the nature of the roots:
   - If $(b^2 - 4ac) > 0$, there are two distinct real roots.
   - If $(b^2 - 4ac) = 0$, there is exactly one real root (a repeated root).
   - If $(b^2 - 4ac) < 0$, there are no real roots (two complex roots).

Classroom Activity

Title: "Quadratic Equation Challenge"

Objective: To practice identifying coefficients and solving quadratic equations using the quadratic formula.

Instructions:

1. Divide the class into small groups (3-4 students per group).
2. Provide each group with a set of three different quadratic equations. Examples:
   - Equation 1: $x^2 - 5x + 6 = 0$
   - Equation 2: $2x^2 + 3x - 5 = 0$
   - Equation 3: $x^2 + 4x + 4 = 0$
3. For each equation, students must:
   - Identify the values of 'a', 'b', and 'c'.
   - Calculate the discriminant $(b^2 - 4ac)$ and state the nature of the roots.
   - Solve the equation using the quadratic formula to find the roots.
4. Groups will present their solutions and discuss any challenges they faced or observations they made about the different types of roots. Encourage them to check their answers by substituting the roots back into the original equation.

Homework

1. Identify the values of a, b, and c for the following quadratic equations:
   a) $3x^2 - 7x + 2 = 0$
   b) $x^2 + 9x = 0$
   c) $5 - 2x + x^2 = 0$
2. Solve the following quadratic equation by factorization:
   $x^2 - 8x + 15 = 0$
3. Use the quadratic formula to find the roots of the equation:
   $2x^2 + 5x - 3 = 0$

ENG

12:32 PM

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Thank you